

Benchmark for Crack Re-Identification in Ticket Management Systems

Délcio Ferramenta, Cátia Loureiro, António Silva, Beatriz Miranda, Paulo Brito, Nuno da Costa, Ricardo Rodrigues, and Du
DTx Colab, Campus de Azurém, Universidade do Minho, 4800-058 Braga, Portugal

Abstract—Building ticket management systems must determine whether a newly reported defect corresponds to an existing ticket or requires a new one. Repeated reports of the same crack from different viewpoints fragment history and duplicate work. We study this association as crack re-identification and introduce CRACKID (140 photographs, 17 walls, 212 identities in the frozen benchmark snapshot: 6 validation walls and 11 test walls with 19 matchable test identities). We evaluate ten zero-shot crop-based matchers from four families under retrieval, thresholded decisions, and open-set identification. Retrieval varies (Rank-1 0.370–0.613), but validation-calibrated pairwise decisions cluster in a narrow band just above the best-constant floor (F_1^{cv} 0.331–0.338 against a 0.305 floor; oracle F_1^* 0.337–0.374). Context and crack-erasure controls implicate wall location over crack appearance. A geometric baseline registering wall texture with cracks masked out and matching projected masks reaches 0.768 detection-and-identification rate at 0.1 false alarms on 48% of pairs, against 0.384 for the best crop method, 0.404 for the best context control, and 0.000 for a coverage-only control. All figures are point estimates on a small wall-clustered sample with single-annotator homography-propagated labels, so the open-set advantage is provenance-conditioned and ranking is short-baseline within-visit correspondence. This establishes a dataset and benchmark for the visual association step, leaving cross-visit performance and ticket reduction for future work.

Index Terms—crack re-identification, dataset, benchmark, building maintenance, structural inspection, homography, geometric alignment, annotation quality

I. INTRODUCTION

When ticket management systems are built, reports of physical defects are arranged into records which can be used for inspection, prioritization, intervention, and follow-up. Each ticket should contain the history of a single incident; yet the same crack may be photographed more than once by different users during various inspections, leading to the creation of duplicate tickets that disrupt the history and result in duplicate assessment work.

To address this, each new observation must be linked to the existing ticket, or a new ticket must be created for any damage not yet recorded.

This is a visual re-identification problem for photographic evidence. There are two cases to consider: (i) the morphology remaining unchanged even when the viewpoint, distance, field of view, illumination, or partial visibility varies; and (ii) physical damage that has changed (by extending, widening, or branching) yet whose identity must still be retained in the face of such changes, to distinguish between damage that has

changed and damage that has been acquired. An error in either case results in a duplicate or misattribution of the observation.

Current methods for detecting cracks focus on detection and segmentation within individual images. Since datasets like CrackSeg9k [1] and OmniCrack30k [2] do not include cross-observation identities, it is not possible to associate an observation with a ticket. The applicability of person/object re-identification [3], [4] to cracks—given that different defects may look similar and one defect can vary greatly—has not been tested.

We developed CRACKID to determine whether photographs show the same wall crack: it includes 140 photos of 17 walls taken at one time (6 for validation and 11 for testing; 19 test identities can be matched, with $n_Q=305$). Real photos account for variations in how images were acquired during a single visit; we initially use a synthetic approach to investigate changes in appearance and growth. Improving performance across visits using real photos will have to be left to future work. The accuracy figures are point estimates based on a small sample of clustered walls.

The benchmark evaluates ten different matchers in a zero-shot setup on CRACKID, separating the process of retrieving candidates from the thresholding step involved in deciding whether cracks are the same, and makes use of open-set identification with rejection: merely ranking a record highly does not mean that a report should be attached to it. By combining context with crack-erased control samples, it is possible to distinguish signals that indicate cracks from those that indicate wall position, and a geometric baseline is used to detect walls and to compare the projected masks.

The study examines RQ1 (the association under acquisition change), RQ2 (the roles of context and geometric alignment), and RQ3 (behavior under synthetic appearance/growth change); for RQ1 and RQ2, real single-visit photos are used, while RQ3 is an exploratory synthetic study. We will address duplicate reduction at the ticket level and cross-visit evaluation in future work.

Contributions: (i) the CRACKID method together with cross-observation identities and information about the source of the annotations (Section III-B); (ii) a benchmark that divides the tasks among retrieval, making same-defect decisions, and open-set identification (Sections IV and VI); (iii) the use of context controls and a geometric reference together with an analysis of coverage (Sections V, VI-C, and VII-B); (iv) an initial synthetic revisit probe (Section VII-D).

II. RELATED WORK

The detection and segmentation of cracks is the focus of crack analysis [5]–[7]; the datasets used in this area also deal with single images, and the consolidation studies follow the same scope [1], [2], and multi-view pipelines combine one inspection into a surface or a segmentation [8] without establishing instance correspondence. Structural-health monitoring compares cracks over time with related equipment [9]–[12], but the identity of the cracks is *given by construction* (using a fixed camera and a marked location): instead of measuring a known crack, they decide on associations under an open-set protocol, which is our aim.

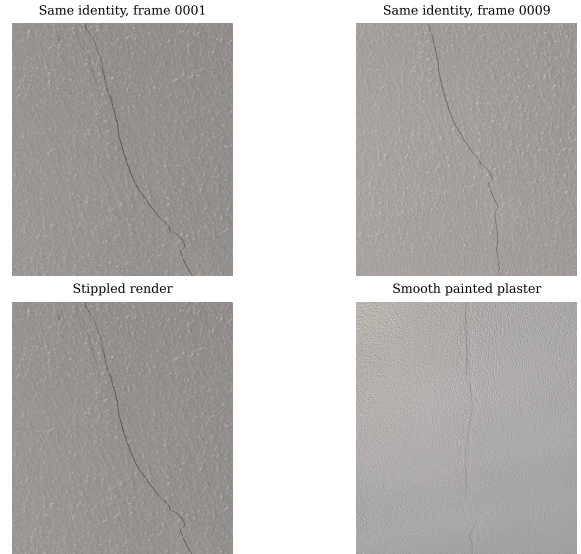
The setting we use is object re-identification [4] together with open-set identification [13]; keypoint matching methods such as SIFT [14] and ORB [15], learned correspondences (including SuperGlue [16] and LoFTR [17]), and region embeddings [3], [18]–[21] can all be applied to crack detection, with all of them being evaluated here in a zero-shot manner without any CRACKID fine-tuning (for the YOLOv8 row: layer-9 features from `yolov8n.pt` [22]; LightGlue [23] and NetVLAD [24] are left as possible retrieval methods for future work). Vehicle-damage re-identification across claims [25] follows the same open-set protocol as ours; the other work in this area focuses on detection and severity rather than on association [26]–[28]. The process of aligning the images and then reasoning about presence and absence constitutes change detection [29], [30] using classical planar registration [31]–[33] and an ECC fallback [34]; our key point is that the alignment and matching targets should be *disjoint*, the homography being determined by the wall since it is the crack that is to be identified.

III. DATA ACQUISITION

A. Acquisition and composition

The CRACKID frozen benchmark snapshot consists of 140 photographs of 17 exterior walls taken at a single location over four half-days using two consumer handheld devices (the data is available through Mendeley Data [35]). The walls have both stippled render (textured) and smooth painted plaster; for the smooth plaster, the only visible features are cracks. The dataset is split into six validation walls (walls 01 to 06; 66 photos, 66 identities) and eleven test walls (walls 07 to 17; 74 photos, 146 identities). We consider only identities that appear in at least two photographs as answerable queries (18 in the validation set and 19 in the test set; $n_Q=305$ in the test set and 439 in the validation set). Additionally, we included two further walls (walls 18 and 19) and therefore omit them from all figures in this paper. Each wall is recorded by a continuous walk-around, with a median interval of 3 seconds between consecutive frames and a maximum duration of 81 seconds; as a result, the data includes within-visit viewpoint and scale correspondence, which is regarded as an upper bound on multi-visit performance.

Adjacent-frame retrieval. For the 305 test queries that have an answer, we use the adjacent frame as the starting point



The images shown here are real CrackID images; the top one shows a single identity spanning non-adjacent frames, while the bottom one shows the stippled render alongside the smooth plaster. We obtained the labels via homography propagation and have not independently audited them.

for the retrieval process. As the gaps shown in Section VII-B (ranging from 305 down to 205, then to 149 and finally to 105 for the gaps numbered 0 to 3) make clear, Rank-1 is still achieved as the number of items decreases. This result shows that the method is robust to short-baseline variations rather than indicating a lack of relevance.

B. Annotation, and why it needs auditing

The ORB homography method operates one-to-one, so for each crack, a single click passes from one frame to the next, and the fragment then combines the segments that constitute the individual physical crack. The greatest risk of an incorrect merge is at this stage, such as changing a hard negative into a free positive. Although the number of merges is provided, their accuracy is not. More precisely, 95 of the 212 identities (45%; 70 out of 146, 48% in the test set) result from merges, with median bridged gaps of 228 pixels in the test set and 286 pixels in the validation set. One annotator examined all clicks and thoroughly triaged the unclaimed points (107 out of 1,023 were classified as no crack; full details are in the supplement). Single-annotator triage reduces but does not eliminate merge risk. The figures refer to the frozen snapshot (Table I); the live `dataset/labels/` directory has since been updated (walls 18 and 19 have been added and the changelog has been rebuilt without the gap fields). It will not display these figures until you rerun the benchmark.

IV. BENCHMARK PROTOCOL

The benchmark assesses visual association by verifying, in the case of a new crack observation, whether it matches one of the previously recorded defects or is part of the reference set. This is so because the annotated identities act as a stand-in for

TABLE I
COUNTING UNITS FOR EVERY METRIC (FROZEN SNAPSHOT, $n_Q=305$
TEST).

	Validation	Test
Walls / photographs / identities	6 / 66 / 66	11 / 74 / 146
Identities in ≥ 2 photographs	18	19
Answerable queries	439	305
Positive query-gallery pairs	26 012	5 722
Valid query-gallery pairs	89 118	31 822
Same-wall image pairs	394	301
Scored image pairs (with valid cells)	341	255

the real permanent defect records, the measure thus indicating the accuracy of association rather than that of duplicate-ticket reduction or overall workflow performance.

A. Task formulation and candidate records

A single physical wall is what is meant by a wall; an identity refers to the association of observations concerning the same crack (as described in Sec. III-B). A query relates to a single crack instance; its gallery consists of other instances from photographs of the same wall, on the condition that the wall is known. This approach does not include cross-wall search. A query is considered known if its gallery contains the same identity; if not, it is unknown. Each method gives scores to the query-gallery pairs in order to carry out ranking, thresholding decisions, and identification with rejection. The unit used in the case of pairs is a query-gallery pair; an image pair is the unit for registration or assignment. The Hungarian one-to-one constraint applies only to each image pair; more than one observation can refer to the same defect record.

The numbers of the tests and how common they are are presented in Table I for the frozen benchmark snapshot (this covers 6 validation walls, namely walls numbered 01 to 06; 11 test walls, namely walls numbered 07 to 17; the walls labeled wall18 and wall19, which were added later, are omitted from all the figures). Registration is performed for all 301 pairs of test images that are identical in wall; 255 of these result in scored pairs with validity-mask cells (for further details, refer to Sec. VII-B).

B. Unchanged damage under different acquisition conditions

The main evaluation involves real walk-around inspections on a per-wall basis, with morphology assumed essentially unchanged and differences arising from viewpoint, distance, scale, lighting, and visibility, along with same-wall distractors and unknown queries. The validation and test phases do not share walls. Each method is used as-is, without fine-tuning with CRACKID on shared masks (Sec. V); thresholds are set on the validation data and then applied to the test data. This establishes within-visit correspondence, but it does not address cross-visit or progression robustness.

C. Controlled appearance change and damage progression

A synthetic reanalysis separates brightness variations from those caused by structural growth by using ten photographs

from five walls, along with one initial seed, and applying the same slight affine warp in each case. In condition B, the appearance is degraded (with blur, noise, and changes in contrast or brightness) even though the structure remains unchanged; in condition A, one end of the largest component is extended. The code specifies all parameters. The study compares OSNet with context to the HYBRID-SKEL method (which combines geometric alignment with a fallback/blend based on the skeleton, $\tau=8.0$ px), which differs from the main REG+CHAMFER reference. Because we use only one seed and five walls, the results are preliminary checks, not longitudinal evidence.

D. Decision rules and handling of unavailable scores

When a pair meets the threshold, it is accepted; at the query level, if the threshold is met, then only the top entry from the gallery is accepted; otherwise no acceptance occurs; in the case of assignment, the Hungarian algorithm is applied to each image block and any assignments below the validation threshold are discarded. Unscorable pairs are retained: for the pairwise and assignment F1 scores, a predicted negative is used, and in retrieval the pair is given the last rank. Thus, it is possible to distinguish between failing to find a known query and correctly rejecting an unknown query and a supported non-match; since a registration failure is not resolved and may need further examination, it should not be treated as evidence of a new defect. For the assignment F1 score, which equals $2TP$ divided by $(2TP + FP + FN)$, correctly accepted assignments are true positives, incorrectly accepted assignments are false positives, and known instances that are not matched are false negatives.

E. Evaluation metrics and reference baselines

We assess the existing queries using Rank-1, Rank-5, and mAP; pairwise decisions are made on the basis of precision, recall and the F_1 measure, with the oracle F_1^* being separated from the validation-calibrated $F_1^{@v}$. We employ F_1^a when dealing with assignments. In the case of identification involving rejection, DIR (based on known accepted-correct examples) and FAR (for unknown accepted cases) are used; the constrained operating point is given by $DIR@FAR=0.1$, which is different from that achieved with a fixed threshold. Coverage is the proportion of scored pairs and is presented together with accuracy. The references consist of random ranking and the best-constant pairwise F_1 of $2p/(1+p)$ (0.305 at the test prevalence of 0.180; 0.544 on the 0.48 geometric subset; 0.452 at the validation prevalence of 0.292); the complete chance levels are given in Table II. We resample under the shared capture conditions; since there are 11 test walls and 19 matchable identities, we give only point estimates together with wall-level intervals, and we will publish the score matrix in our future work (Sec. VIII).

V. A GEOMETRIC REFERENCE BASELINE

The baseline tests whether crack locations on the wall drive the association: it aligns wall surfaces to a common frame

and then compares the projected crack masks. It assumes a roughly planar wall and uses no CRACKID training. The shared UNet++ masks (trained on public crack data only; no CRACKID wall appears in their training set) use instance extraction at 200,px and 5,px to keep inputs consistent across methods (full segmenter settings are in the code). Masks are taken as given, and their error is not measured.

A. Registration using wall texture

For I_a, I_b , a homography H ($x' \sim Hx$) is estimated from the surrounding wall texture with detected cracks dilated and excluded, so alignment is driven by surface features independently of the identity under test. This independence is conditional on the masks and the stage that succeeds, not guaranteed per pair. Full detector settings (SIFT with CLAHE contrast enhancement [36], ratio test, robust estimation) are in the code.

B. Registration cascade and validity checks

Textured render generally provides background features, whereas smooth plaster does not. A three-stage cascade is employed: first masked SIFT; if too few keypoints remain, unmasked SIFT as a fallback (flagged per pair, since the crack then aids alignment); and finally ECC photometric alignment. Candidates pass plausibility tests (no mirrored or folded quads; scale within $[1/16, 16]$); the surviving stage is recorded per pair, and its inlier RMSE sets the comparison tolerance. Unsourced pairs count as lack of evidence, not evidence of different defects.

C. Crack projection and geometric comparison

Accepted registrations project instances from I_a into I_b for comparison by symmetric mean Chamfer distance, which tolerates the small misalignments that would collapse IoU on narrow cracks. Let $d(p, b) = \min_{q \in b} |p - q|/2$:

$$D(a, b) = \frac{1}{2}! \left[\frac{1}{|a|} \sum p \in a d(p, b); +; \frac{1}{|b|} \sum q \in b d(q, a) \right], \quad (1)$$

Early rejection of distant candidates applies. Chamfer coverage (the fraction within tolerance τ of the per-pair RMSE, clamped to $[2, 16]$ pixels, default 6 pixels; HYBRID-SKEL uses 8.0 pixels) is evaluated separately as its scale-free counterpart (see Sec. VII-C).

D. Correspondence decisions and limits of spatial evidence

For each image pair, a one-to-one assignment follows the Hungarian method [37] with the validation threshold shared by all methods; unsourced pairs take value $-\infty$ (predicted negative, ranked last, as in Sec. IV). An aligned projection onto visibly empty wall is a **licensed negative**—evidence of absence, stronger than low embedding similarity—while missed detections, out-of-frame areas, and registration failures stay unresolved. The REG+CHAMFER reference targets unchanged damage; progression uses HYBRID-SKEL. Performance depends on surface-dependent failures and on the homography-based annotation provenance.

VI. THE APPEARANCE CEILING

A. Setup

The ten methods are assessed using crack-centered crops and a common validity mask: SIFT and ORB together with RANSAC inliers, SuperGlue and LoFTR using pretrained indoor settings, five embedding-based methods, and re-ID OSNet (using cosine similarity, with all being zero-shot on CRACKID); YOLOv8 makes use of the layer-9 `yolo8n.pt` features. The full input sizes and checkpoints are given in the code repository. For the context controls, OSNet and CLIP are used with a fixed backbone: `@ctx1` increases the size of the instance box by $1\times$ in each direction, and `@ctx1@nocrack` fills in the crack in the same view to isolate the wall context. The REG+CHAMFER method aligns the walls and then compares the projected masks (as described in Sec. V); the COVERAGE-ONLY control assigns a constant score to the registered pairs without examining the cracks. In the synthetic protocol, we compare OSNet with context against the different HYBRID-SKEL configurations. LightGlue and NetVLAD are regarded as future retrieval baselines.

B. Result: appearance buys almost nothing on the decision

The primary benchmark results, as shown in Table II (11 walls; frozen snapshot; $n_Q=305$ answerable test queries), show that the Rank-1 performance varies between 0.370 and 0.613 (ViT-B/16 obtains the highest score; OSNet achieves the best mAP at 0.497; SuperGlue gives the best crop $\text{DIR@FAR} = 0.1$ at 0.384). Yet the oracle F_1^* is in the range from 0.337 to 0.374 and the validation-calibrated F_1^{cal} lies between 0.331 and 0.338, which shows that good ranking performance does not imply accurate decision-making. The small range of the validation-calibrated F_1 indicates difficulty transferring a decision threshold across walls. With 19 matchable test identities, we do not claim fine-grained ranking among the crop rows. All the values reported are point estimates.

C. Is it the crack, or the wall behind it?

Contextual information has a positive effect on both ranking and open-set performance, yet even when it is removed the majority of this benefit is still retained: OSNet's Rank-1 score rises from 0.603 to 0.775 (it reaches 0.711 when the context is removed), DIR@FAR increases from 0.154 to 0.404 (it is 0.393 when the context is removed), and CLIP's open-set performance improves from 0.143 to 0.311 (it amounts to 0.286 when the context is removed). The two views with the context removed outperform all methods that use tight crops, indicating that tall location is the main clue. The pairwise F_1^* remains within a narrow range (from 0.307 to 0.361) for the base, context, and erased views, while open-set performance improves by +0.250 for OSNet. Nevertheless, although the methods examined can retrieve within-visit observations, reliable automatic identity decisions are still not possible. A ranked list may help an operator, but acceptance will need more evidence.

TABLE II

MAIN RESULT ON CRACKID TEST (11 WALLS; FROZEN SNAPSHOT, $n_Q=305$). F_1^* IS COMPUTED USING AN ORACLE THRESHOLD ON THE TEST SET; ΔF_1^* REPRESENTS THE EXCESS OVER THE FULL-POOL FLOOR. $F_1^{\text{@v}}$ IS CALCULATED USING THE VALIDATION-CALIBRATED THRESHOLD. BOLD FONT INDICATES THE BEST CROP ROW IN EACH COLUMN. THE GEOMETRIC ROW IS BASED ON A SUBSET OF SIZE 0.48 WITH A FLOOR OF 0.544. THEREFORE, ITS ΔF_1^* IS NOT DIRECTLY COMPARABLE TO THAT OF THE CROP ROWS; SEE SEC. VII-B. FOR DETAILS ON SCORING, CHANCE LEVELS, AND COMPUTATIONAL COSTS, REFER TO SEC. IV THROUGH SEC. VII-B. THE METRIC S/QUERY IS DEFINED AS TOTAL_S/305 ON A SINGLE WORKSTATION (RELATIVE TIMINGS ONLY).

Method	Family	R@1	mAP	DIR@FAR=0.1	F_1^*	ΔF_1^*	$F_1^{\text{@v}}$	scored	s/query
<i>Reference level (same 0.48 pool as geometry)</i>									
COVERAGE-ONLY	geometry	0.515	0.533	0.000	0.544	0.239	0.544	0.48	—
<i>Crop scope</i>									
ORB [15]	keypoint	0.370	0.372	0.043	0.337	0.032	0.337	1.00	0.077
SIFT [14]	keypoint	0.420	0.408	0.049	0.343	0.038	0.337	1.00	0.155
SuperGlue [16]	learned	0.518	0.434	0.384	0.357	0.052	0.337	1.00	5.196
LoFTR [17]	learned	0.512	0.424	0.285	0.357	0.052	0.337	1.00	7.190
YOLOv8 backbone [22]	embedding	0.525	0.439	0.134	0.374	0.069	0.338	1.00	0.016
ViT-B/16 [18]	embedding	0.613	0.487	0.259	0.344	0.039	0.336	1.00	0.021
DeiT-S [19]	embedding	0.570	0.479	0.223	0.337	0.032	0.337	1.00	0.008
CLIP ViT-B/32 [20]	embedding	0.604	0.471	0.143	0.307	0.002	0.306	1.00	0.050
DINOv2 [21]	embedding	0.567	0.459	0.272	0.337	0.032	0.331	1.00	0.011
OSNet [3]	re-ID	0.603	0.497	0.154	0.338	0.033	0.337	1.00	0.006
<i>Full-image scope</i>									
REG+CHAMFER (reference)	geometry	0.952	0.886	0.768	0.656	—	0.582	0.48	27.92

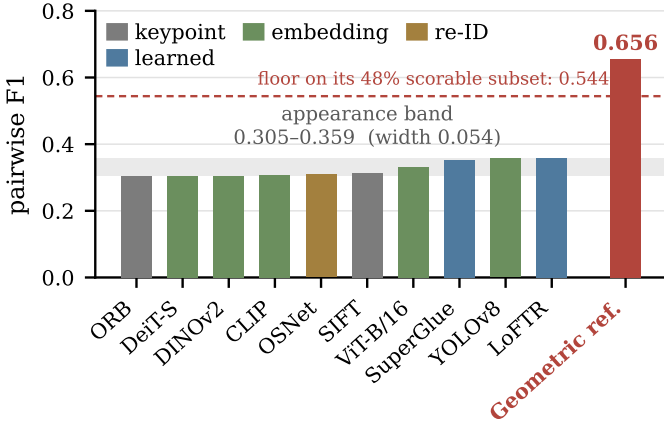


Fig. 1. The appearance ceiling: pairwise F_1^* stays narrow while ranking spreads. Geometry (red) leaves the band but is in the 0.48 pool (own floor 0.544). Table II takes precedence.

VII. RESULTS AND DISCUSSION

A. Main comparison and the open-set gap

Geometric Rank-1 achieves a score of 0.952, whereas the highest score is 0.613 (which is attained by ViT-B/16), and the mAP is 0.886 as compared to 0.497 (that of OSNet) (the score is 0.48 compared to 1.00). When coverage is measured in terms of the amount exceeding COVERAGE-ONLY, the ranking improvements remain substantial, even though the open-set figure is one that coverage does not consider: DIR@FAR=0.1 is 0.768 as against 0.384 (SuperGlue), 0.404 (the best result with OSNet+ctx1), and 0.000 for COVERAGE-ONLY itself (chance DIR is 0.038). For self-similar hairlines, the crop similarities almost always lead to non-rejection, while an aligned

empty projection yields a valid negative. On cost (LoFTR takes 7.19 seconds whereas OSNet takes 0.006 seconds per query; Reg requires 27.9 seconds per query and 26.0 seconds per image pair; the timings are given only relative to each other and SuperGlue/LoFTR were timed in a separate run). As for RQ2 (the geometric aspect): alignment offers identification capabilities that go beyond what coverage alone can provide, on the condition that success, surface, and provenance are taken into account. All the figures are point estimates based on 19 matchable identities.

B. Partial coverage, gap structure, and failures

Crop methods manage to get 100% of the pairs, while geometry only achieves 48%; unscored pairs are given a negative score and are ranked at the end. Censoring has the effect of penalizing the ranking, but it also leads to an improvement in the pairwise F_1 : the group that is scored keeps 100% of the positives but only 36.8% of the negatives (the prevalence increases from 0.180 to 0.373 and the floor rises from 0.305 to 0.544). Since the coverage stays the same for registered pairs in the absence of crack inspection, the coverage is 0.544 for pairwise F_1 , 0.533 for mAP, and 0.515 for Rank-1: a large part of the ranking and pairwise performance therefore depends on *which pairs align*. The only exception is the open-set lead (0.768 compared to 0.000). The failures are due to the wall (zero per cent in the case of the two test walls; the complete per-wall table is provided in the supplement), so the overall scores vary according to surface. The adjacent frame answers all 305 answerable queries, so the main ranking is based on short-baseline correspondence; the gap sweeps (which decrease from 305 to 105 when the gap is 3) keep Rank-1 unchanged even though the pool is shrinking, rather than meaning that the queries are irrelevant.

Protocol	Method	n_q	R@1	mAP	DIR@FAR _{0.1}	$F_1^{@v}$	F_1^a
A (growth) OSNet+ctx	HYBRID-SKEL	69	0.275	0.401	0.343	0.569	0.646
	69	0.826	0.647	0.739	0.601	0.631	
B (appear.) OSNet+ctx	HYBRID-SKEL	82	0.274	0.403	0.342	0.574	0.640
	82	0.671	0.570	0.663	0.594	0.609	

TABLE III

HYBRID-SKEL VS OSNet+CTX ON SYNTHETIC REVISIT; F_1 AT VALIDATION THRESHOLD. ONE SEED, FIVE WALLS, NO RESAMPLING; CHECKPOINT, NOT STABILITY CLAIM. HYBRID-SKEL $\neq$ TABLE II REG+CHAMFER.

Registration is based on image sharpness rather than on frame gap (wall-clustered Wald $z = 5.11$, $p = 3.3 \times 10^{-7}$ versus -2.78 , $p = 5.4 \times 10^{-3}$; further details are given in the supplement); using a single sharpness threshold (≥ 10) increases registration to 56.1% on the filtered set at the expense of one wall, demonstrating that guidance based on capture quality is more appropriate than full-population coverage.

C. Ablations and cost

The cost of using Coverage (scale-free) is 0.240 mAP and 0.103 Rank-1 compared to that of Chamfer distance, with DIR remaining almost the same at $+0.014$; the comparison rule affects the ranking rather than the decisions. When crack exclusion is disabled, the ranking and pairwise results are unchanged, but DIR increases by $+0.057$; masking is kept in order to ensure evidence independence, even though observed accuracy is not discussed here. ECC contributes $\sim +0.03$ Rank-1 and $+0.02$ mAP with no change to DIR. The time required is 26.0 seconds per image pair (7817.3 seconds over 301 test pairs, that is 27.9 seconds per query as shown in Table II; Chamfer-only afterward), which is about $3\times$ the time of LoFTR per query and more than two orders of magnitude higher than that of the cheapest embedding; the absolute timings depend on the workstation, and the times for SuperGlue and LoFTR were obtained from a separate run.

D. A synthetic revisit protocol

Ten source photographs taken from five walls (with one seed) are distorted in the same way using either method (A) single-tip growth or (B) photometric corruption and are then compared with the actual images from those same walls (the full procedure is given in the code; the degree of distortion is small, and the $p90$ tilt is 75°). HYBRID-SKEL differs from REG+CHAMFER as indicated in Table II; consequently, the entries in the rows represent a controlled variation rather than the main configuration.

In the ranking/open-set cases OSNet+ctx does better (the figures being 0.826/0.671 and 0.739/0.663 as compared to 0.275/0.274 and 0.343/0.342), while HYBRID-SKEL achieves a higher assignment F_1^a (0.646/0.640 compared to 0.631/0.609) and the pairwise $F_1^{@v}$ values are very similar (0.569/0.574 versus 0.601/0.594); no assignment cell came from pure homography alone ($H=0.0\%$). Ordering thus reverses with the metric (RQ3): the Table II advantage does not

imply multi-visit robustness, and single-seed controlled growth does not establish real progression performance.

VIII. VALIDITY, LIMITATIONS, AND FUTURE WORK

This study provides the first measurable approach to re-identifying cracks in a single visit and establishes a geometric reference standard. Because the results are based on a small sample of wall-clustered data with annotations from a single annotator and only partial registration coverage, each finding below is accompanied by the specific step needed to verify or build on it.

- 1) **Scale the system to handle data from multiple walls and multiple visits.** Gather information from more walls, more surface families, and more annotators to establish confidence intervals at the wall level and to enable reliable ranking of the methods. Most importantly, include a second capture session at deliberately different distances and lighting conditions to turn the within-visit measurement into a true cross-visit benchmark for ticket management backends.
 - 2) **Add annotation that does not depend on provenance.** Although a single annotator verified each click and also triaged any unclaimed points, we have not measured agreement, and the geometric advantage still depends on provenance. By including a small subset of labels obtained entirely without using homography propagation, together with the release of per-pair score matrices for use in sensitivity analysis and for wall-clustered intervals and threshold-free AP/AUROC, it would be possible to determine how much of the gap in the open-set case is independent of the provenance of the annotation.
 - 3) **Close the gap in registration coverage.** The baseline responds to 48 percent of the pairs (97 out of 301 image pairs; 15,328 out of 31,822 valid cells) and gives a score of zero on smooth-plaster walls. Registration that is adaptive to the surface—this involving the learning of local features, direct photometric alignment, or sparse depth obtained from consumer devices—would move coverage towards complete evaluation and also further separate the A geometry pipeline based on homography-based provenance.
 - 4) **Learn from supervision derived from geometry.** The baseline is not a system that has been deployed but rather a reference that is not learned. If the registration pipeline generates pseudo-labels, or if a foundation-model backbone is fine-tuned using homography-derived correspondences, the location-based advantage will be retained in pair registration.
- Assess the ticket lifecycle by defining task-level metrics—specifically the duplicate rate and the missed-new-defect rate—and then examine the effects of segmentation, re-identification, and joint ticket assignment in a closed-loop simulation of actual inspection procedures.

Taken together, these steps convert the current within-visit reference into a deployable A component which can be used

for re-identification across multiple visits to handle operational tickets.

IX. CONCLUSION

We have developed a dataset that enables the testing of crack re-identification: CrackID, which includes 140 photographs taken from 17 walls and covering 212 identities in the frozen snapshot (6 of these being validation walls and 11 being test walls; walls 18 and 19 were excluded); the source of the annotations (45% of which were merged and 48% of which came from the test set) is stated rather than assumed, and the procedure for handling unclaimed clicks has now been completed (in 1,023 cases, 107 did not involve the naming of a crack). All of the accuracy figures are point estimates and do not feature wall-level confidence intervals (19 test identities were matchable, with $n_Q=305$); we are asserting the pattern, not the individual digits.

The key result of the benchmark is negative and structural in nature: when a decision is required, ten of the zero-shot matchers from four different families achieve a score between +0.032 and +0.069 above the 0.305 best-constant floor (F_1^* ranging from 0.337 to 0.374), while validation-calibrated decisions cluster in a narrow band just above it (F_1^{cv} ranging from 0.331 to 0.338). Determining a person's identity from a gallery is separate from making a decision. The problem does not lie merely in the appearance of cracks: a geometric reference baseline uses wall texture as a reference while masking the cracks. Chamfer-matches are projected achieves a DIR@FAR=0.1 of 0.768 compared to 0.384 for the best crop method and 0.404 for the best context method, correctly identifying 48% of the pairs, and a COVERAGE-ONLY control that makes use of that alignment without looking at the crack itself obtains a score of 0.000 on the open-set metric (although it has a pairwise F_1 of 0.544 on its own 0.544 floor).

As far as operation is concerned, the advantage in the open-set scenario results from an aligned pair serving as a *licensed negative*, that is, positive evidence of absence, since a crop embedding can only show low similarity; it is still not clear whether this benefit is independent of homography-based annotation, and it must therefore be regarded as depending on the source. Two issues remain: a real second capture session changes the benchmark from viewpoint-based to cross-visit, and adding a second annotator with a homography-free audit subset would turn single-annotator triage into a measured agreement rate. The 48% coverage is also, on its own, a deployment shortfall, not merely a scoring issue: smooth-plaster walls can be categorically unanswerable. Finally, the small single-seed synthetic revisit checkpoint reverses the ranking referred to in the headline and gives the open-set result in favor of OSNet+context, while the distinct HYBRID-SKEL pipeline retains only a slight advantage in assignment- F_1 ; the paper in question therefore offers a promising within-visit geometric reference, rather than solving the re-identification aspect for a multi-visit ticket management backend.

ACKNOWLEDGMENT

This work was supported by “R2UTechnologies — modular systems” (C644876810-00000019), by “R2UTechnologies” Consortium, cofinanced by NextGenerationEU, through the Incentive System “Agendas para a Inovação Empresarial” (“Agendas for Business Innovation”), within the Recovery and Resilience Plan (PRR).

REFERENCES

- [1] S. Kulkarni, S. Singh, D. Balakrishnan, S. Sharma, S. Devunuri, and S. C. R. Korpapati, “CrackSeg9k: A collection and benchmark for crack segmentation datasets and frameworks,” 2022.
- [2] C. Benz and V. Rodehorst, “OmniCrack30k: A benchmark for crack segmentation and the reasonable effectiveness of transfer learning,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2024.
- [3] K. Zhou, Y. Yang, A. Cavallaro, and T. Xiang, “Omni-scale feature learning for person re-identification,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 3702–3712.
- [4] Y. Qian, J. Barthelemy, E. Karupiah *et al.*, “Identifying re-identification challenges: Past, current and future trends,” *SN Computer Science*, vol. 5, p. 937, 2024.
- [5] Q. Zou, Y. Cao, Q. Li, Q. Mao, and S. Wang, “CrackTree: Automatic crack detection from pavement images,” *Pattern Recognition Letters*, vol. 33, no. 3, pp. 227–238, 2012.
- [6] Y. Liu, J. Yao, X. Lu, R. Xie, and L. Li, “DeepCrack: A deep hierarchical feature learning architecture for crack segmentation,” *Neurocomputing*, vol. 338, pp. 139–153, 2019.
- [7] M. Eisenbach, R. Stricker, D. Seichter, K. Amende, K. Debes, M. Sessmann, D. Ebersbach, U. Stoeckert, and H.-M. Gross, “How to get pavement distress detection ready for deep learning? a systematic approach,” in *Int. Joint Conf. Neural Networks (IJCNN)*, 2017, pp. 2039–2047.
- [8] C. Benz and V. Rodehorst, “CrackStructures and CrackEnsembles: The power of multi-view for 2.5d crack detection,” in *Proceedings of the Winter Conference on Applications of Computer Vision (WACV)*, 2025, pp. 5990–5999.
- [9] L. Parente, E. Falvo, C. Castagnetti, F. Grassi, F. Mancini, P. Rossi, and A. Capra, “Image-based monitoring of cracks: Effectiveness analysis of an open-source machine learning-assisted procedure,” *Journal of Imaging*, vol. 8, no. 2, p. 22, 2022.
- [10] S. Harb, P. Achancaray, M. Maboudi, and M. Gerke, “Multi-temporal crack segmentation in concrete structures using deep learning approaches,” *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. X-G-2025, pp. 341–348, 2025.
- [11] X. Sun and P. Chang, “Robust perspective correction for real-world crack evolution tracking in image-based structural health monitoring,” 2025.
- [12] NEC Corporation, “Photographing guide device,” US Patent 12,223,639 B2 (priority 2019), 2025.
- [13] S. Liao, Z. Mo, Y. Hu, and S. Z. Li, “Open-set person re-identification,” arXiv preprint arXiv:1408.0872, 2014.
- [14] D. G. Lowe, “Distinctive image features from scale-invariant keypoints,” *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004.
- [15] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, “ORB: An efficient alternative to SIFT or SURF,” in *IEEE International Conference on Computer Vision (ICCV)*, 2011, pp. 2564–2571.
- [16] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich, “SuperGlue: Learning feature matching with graph neural networks,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 4938–4947.
- [17] J. Sun, Z. Shen, Y. Wang, H. Bao, and X. Zhou, “LoFTR: Detector-free local feature matching with transformers,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 8922–8931.
- [18] A. Dosovitskiy, L. Beyer, A. Kolesnikov *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *International Conference on Learning Representations (ICLR)*, 2021.

- [19] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers and distillation through attention," in *International Conference on Machine Learning (ICML)*, 2021, pp. 10 347–10 357.
- [20] A. Radford, J. W. Kim, C. Hallacy *et al.*, "Learning transferable visual models from natural language supervision," in *International Conference on Machine Learning (ICML)*, 2021, pp. 8748–8763.
- [21] M. Oquab, T. Darcet, T. Moutakanni *et al.*, "DINOv2: Learning robust visual features without supervision," *Transactions on Machine Learning Research*, 2024.
- [22] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," <https://github.com/ultralytics/ultralytics>, 2023.
- [23] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys, "LightGlue: Local feature matching at light speed," in *IEEE/CVF Int. Conference on Computer Vision (ICCV)*, 2023, pp. 17 627–17 638.
- [24] R. Arandjelović, P. Gronat, A. Torii, T. Pajdla, and J. Sivic, "NetVLAD: CNN architecture for weakly supervised place recognition," in *IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 5297–5307.
- [25] L. Maiano, A. Montuschi, M. Caserio *et al.*, "A deep-learning-based antifraud system for car-insurance claims," *Expert Systems with Applications*, vol. 231, p. 120644, 2023.
- [26] M. J. Hasan, C. K. Nguyen, Y. L. Boo, H. Jahani, and K.-L. Ong, "Vehicle damage detection using artificial intelligence: A systematic literature review," *WIREs Data Mining and Knowledge Discovery*, vol. 15, p. e70027, 2025.
- [27] S. A. Pérez-Zarate, D. Corzo-García, J. L. Pro-Martín *et al.*, "Automated car damage assessment using computer vision: Insurance company use case," *Applied Sciences*, vol. 14, no. 20, p. 9560, 2024.
- [28] D. Mallios, L. Xiaofei, N. McLaughlin, J. M. Del Rincon, C. Galbraith, and R. Garland, "Vehicle damage severity estimation for insurance operations using in-the-wild mobile images," *IEEE Access*, vol. 11, pp. 78 644–78 655, 2023.
- [29] K. Sakurada and T. Okatani, "Change detection from a street image pair using CNN features and superpixel segmentation," in *British Machine Vision Conference (BMVC)*, 2015.
- [30] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, "Street-view change detection with deconvolutional networks," *Autonomous Robots*, vol. 42, no. 7, pp. 1301–1322, 2018.
- [31] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2004.
- [32] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," in *Readings in Computer Vision*. Morgan Kaufmann, 1987, pp. 726–740.
- [33] D. Barath, J. Matas, and J. Noskova, "MAGSAC: Marginalizing sample consensus," in *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 10 197–10 205.
- [34] G. D. Evangelidis and E. Z. Psarakis, "Parametric image alignment using enhanced correlation coefficient maximization," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 30, no. 10, pp. 1858–1865, 2008.
- [35] D. Ferramenta, "CrackID," Mendeley Data, Version 1, <https://doi.org/10.17632/t749fc4x44.1>, 2026.
- [36] K. Zuiderveld, "Contrast limited adaptive histogram equalization," in *Graphics Gems IV*. Academic Press, 1994, pp. 474–485.
- [37] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1-2, pp. 83–97, 1955.